

1. Title

ShopStack : Enterprise Multi-Vendor E-Commerce Platform

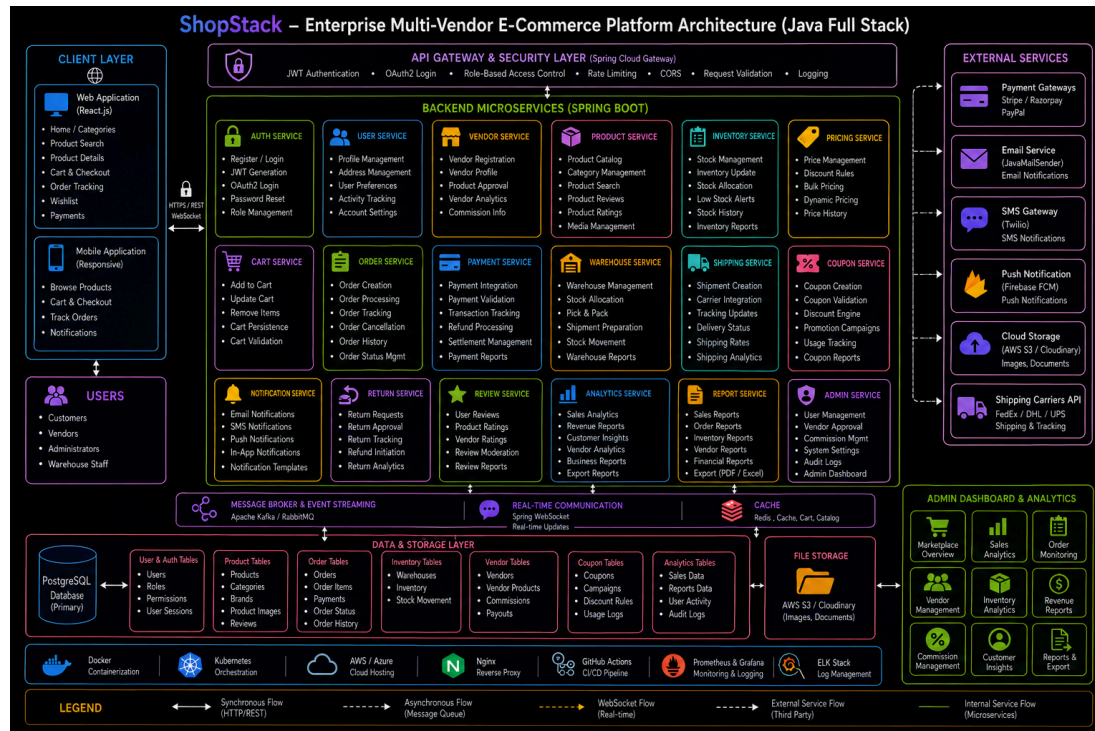
2. Objective

Build a full-stack web application (React.js frontend + Spring Boot backend) that enables customers to purchase products from multiple vendors through a centralized marketplace. The platform provides vendor onboarding, product management, inventory tracking, secure payments, order processing, warehouse management, coupon management, shipment tracking, and analytics dashboards. This solution can be used for enterprise marketplaces, retail businesses, and large-scale commerce platforms.

Outcomes

- (i) Developed and deployed a scalable enterprise multi-vendor e-commerce platform.
- (ii) Implemented customer shopping, cart, wishlist, and secure payment workflows.
- (iii) Built vendor product, pricing, and inventory management modules.
- (iv) Developed warehouse allocation and shipment tracking systems.
- (v) Implemented coupon engine and promotional discount workflows.
- (vi) Built admin dashboard for vendor approvals, analytics, and commission management.
- (vii) Developed order processing, return, and refund management systems.
- (viii) Deployed the application using Docker and cloud platforms.
- (ix) Added reporting and business analytics modules.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login
- Password Reset
- Profile Management
- Role Management

Roles

- Customer
- Vendor
- Administrator
- Warehouse Staff

2. Customer Module

- (i) Customer registration
- (ii) Product browsing
- (iii) Product search
- (iv) Wishlist management
- (v) Cart management
- (vi) Order history
- (vii) Profile management

3. Vendor Module

- (i) Vendor registration
- (ii) Product listing
- (iii) Inventory management
- (iv) Price management
- (v) Sales monitoring
- (vi) Vendor analytics
- (vii) Product approval workflow

4. Product Catalog Module

- (i) Product management
- (ii) Category management
- (iii) Product search
- (iv) Product filtering
- (v) Product reviews

(vi) Product ratings

Product Information

- Product Name
- Category
- Brand
- Description
- Images
- Price
- Stock Quantity
- Vendor Details

5. Inventory Management Module

(i) Stock tracking

(ii) Inventory updates

(iii) Inventory allocation

(iv) Inventory reports

(v) Low stock alerts

(vi) Stock history tracking

6. Cart & Checkout Module

(i) Add to cart

(ii) Remove from cart

(iii) Update quantities

(iv) Checkout workflow

(v) Address management

(vi) Order confirmation

7. Payment Module

- (i) Online payments
- (ii) Payment verification
- (iii) Refund processing
- (iv) Transaction history
- (v) Payment status monitoring
- (vi) Settlement workflows

8. Order Management Module

- (i) Order creation
- (ii) Order processing
- (iii) Order tracking
- (iv) Order cancellation
- (v) Return requests
- (vi) Refund workflows

Order Status

- Pending
- Confirmed
- Processing
- Shipped
- Delivered
- Returned
- Refunded

9. Warehouse Management Module

- (i) Warehouse allocation
- (ii) Pick and pack operations
- (iii) Inventory synchronization

- (iv) Warehouse analytics
- (v) Stock movement tracking
- (vi) Shipment preparation

10. Shipping & Tracking Module

- (i) Shipment creation
- (ii) Shipment tracking
- (iii) Delivery monitoring
- (iv) Tracking updates
- (v) Delivery notifications
- (vi) Shipping analytics

11. Coupon & Promotion Engine

- (i) Coupon creation
- (ii) Discount management
- (iii) Promotional campaigns
- (iv) Offer validation
- (v) Coupon tracking
- (vi) Coupon analytics

12. Notification Module

- (i) Order updates
- (ii) Payment notifications
- (iii) Shipment alerts
- (iv) Vendor notifications

(v) Email notifications

(vi) SMS notifications

(vii) Push notifications

13. Analytics Dashboard Module

Customer Dashboard

(i) Active orders

(ii) Order history

(iii) Wishlist overview

(iv) Personalized recommendations

(v) Account activity

Vendor Dashboard

(i) Sales analytics

(ii) Inventory reports

(iii) Revenue insights

(iv) Product performance

(v) Customer trends

Admin Dashboard

(i) Vendor management

(ii) Marketplace analytics

(iii) Order monitoring

(iv) Commission management

(v) System monitoring

(vi) Business reports

14. Reports & Export Module

(i) Sales reports

(ii) Inventory reports

(iii) Vendor reports

(iv) Order reports

(v) Financial reports

(vi) PDF export

(vii) Excel export

15. Final Integration, Testing & Deployment

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

(i) Define marketplace objectives and business workflows.

(ii) Design system architecture and database schema.

(iii) Create UI wireframes and marketplace flow planning.

(iv) Setup React frontend and Spring Boot backend.

(v) Implement authentication and role management.

(vi) Build customer and vendor management modules.

(vii) Develop product catalog management workflows.

Outcomes

(i) Understand enterprise e-commerce workflows and marketplace systems.

(ii) Learn system architecture and database design concepts.

(iii) Build frontend and backend project initialization.

(iv) Working authentication and product management system.

Milestone 2: Week 3 & 4 — Inventory, Pricing & Order Processing

Tasks

(i) Implement inventory tracking and stock management.

(ii) Build pricing and discount management workflows.

(iii) Develop cart and checkout functionality.

(iv) Integrate secure payment gateway.

(v) Implement order tracking and status updates.

Outcomes

(i) Implement inventory and pricing management systems.

(ii) Build secure order and payment processing workflows.

(iii) Understand enterprise commerce transaction handling.

(iv) Generate real-time order tracking functionality.

Milestone 3: Week 5 & 6 — Admin Management, Coupon Engine & Warehouse Logic

Tasks

- (i) Build admin dashboard and reporting modules.
- (ii) Implement vendor commission calculations.
- (iii) Develop coupon and promotion engine.
- (iv) Build warehouse allocation workflows.
- (v) Implement return and refund management.

Outcomes

- (i) Build enterprise-level admin and reporting systems.
- (ii) Understand warehouse and logistics workflows.
- (iii) Implement coupon and promotional management.
- (iv) Complete end-to-end order processing lifecycle.

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform system testing and bug fixing.
- (ii) Improve UI responsiveness and error handling.
- (iii) Deploy application using Docker and cloud platforms.
- (iv) Prepare final project documentation and presentation.
- (v) Demonstrate the complete e-commerce workflow.

Outcomes

- (i) Gain deployment and testing experience.
- (ii) Improve application stability and usability.
- (iii) Complete live deployment and final demonstration.
- (iv) Prepare professional project documentation and presentation.

6. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Project initialization and architecture setup completed.
- (ii) Authentication and user management implemented.
- (iii) Product browsing and vendor management functional.
- (iv) System design and UI planning completed.

Milestone 2 (Week 4)

- (i) Inventory and pricing workflows implemented.
- (ii) Cart and checkout functionality working.
- (iii) Payment gateway integration completed.
- (iv) Order tracking workflows functional.

Milestone 3 (Week 6)

- (i) Admin dashboard and commission system implemented.
- (ii) Coupon engine and promotional workflows functional.
- (iii) Warehouse allocation and shipment processing working.
- (iv) Return and refund workflows integrated.

Milestone 4 (Week 8)

- (i) Fully deployed frontend and backend.
- (ii) Testing and validation completed.
- (iii) Documentation and presentation prepared.
- (iv) Successful end-to-end project demonstration completed.

7. Tools & Tech Stack

Programming Language

Backend

- (i) Java
- (ii) Spring Boot

Frontend

- (i) JavaScript
- (ii) React.js

Database

Primary Database

- (i) PostgreSQL

Cache

- (i) Redis

Authentication

- (i) Spring Security
- (ii) JWT Authentication
- (iii) OAuth2 Login

Libraries & Frameworks

Backend

- (i) Spring Boot

- (ii) Spring Security
- (iii) Spring Data JPA
- (iv) Hibernate
- (v) Maven
- (vi) Spring WebSocket

Frontend

- (i) React.js
- (ii) React Router
- (iii) Axios
- (iv) Tailwind CSS
- (v) Redux Toolkit
- (vi) Context API

Payment Integration

- (i) Stripe
- (ii) Razorpay
- (iii) PayPal

Storage

- (i) AWS S3
- (ii) Cloudinary

Notifications

- (i) Firebase Cloud Messaging
- (ii) JavaMailSender

(iii) Twilio SMS

Testing

(i) JUnit

(ii) Mockito

(iii) Postman

(iv) React Testing Library

Dev & Deployment Tools

(i) IntelliJ IDEA

(ii) VS Code

(iii) Git & GitHub

(iv) Docker

(v) Docker Compose

(vi) GitHub Actions

(vii) AWS / Azure

(viii) Nginx

(ix) Postman

8. Performance Metrics

Order Processing Performance

(i) Successful order completion rate

(ii) Payment success rate

(iii) Average order processing time

Inventory Accuracy

- (i) Warehouse stock consistency
- (ii) Inventory synchronization accuracy

System Performance

- (i) API response time
- (ii) Concurrent user handling
- (iii) Database query performance
- (iv) Cache hit ratio

9. Example Quantitative Goals

Order Processing

Achieve high order completion and payment success rates.

Inventory Management

Maintain accurate warehouse stock synchronization.

Marketplace Performance

Support multiple concurrent users with stable performance.

Customer Experience

Deliver smooth shopping, checkout, and order tracking workflows.